

CNSA108 OS and Software Fundamentals

Lab: Windows 10/11 Backup Options

Goals:

Explore backup options in Windows 10/11

You should have:

- Basic understanding of Windows 10/11 interface.
- Basic knowledge of storage devices and system files.
- Access to a Windows 10 and a Windows 11 machine (real or virtual).
- External storage (e.g., USB, external hard drive) or network location for backups.
- Admin rights on the machine you're working on.

For each section below, run the process on Windows 10, and then verify if the same process can be performed on Windows 11.

Section 1: Backing Up Files Using File History

1. Goal: Back up personal files using Windows File History on Windows 10.
2. What is Windows File History? Explain the importance of file backups.
Built in windows tool that makes copies of files to save to network or external hard drive. File backups are important to make sure data isn't lost.
3. Use File History to perform a backup of the Documents folder (or other folder of stored classwork) to an externally connected SSD.
4. Explain in detail how you access, setup, and implement File History backups. Note – this is documenting a set of instructions for future use. A single, long paragraph isn't the preferred format. Screen snips are helpful and prove you did the work:
5. Does this function exist on Windows 11? Is the process the same to configure it?
It still exists but windows doesn't recommend it as the first option and it is in the control panel now.

Section 2: Creating a System Image Backup

6. Goal: create a full system image backup of a drive.
7. How are system image backups useful for restoring an entire system after hardware failure or OS corruption?
It saves everything from your computer at the time and allows for a full restoration.
8. Use Windows 10 built-in Backup and Restore process to create a system image stored on your extra SSD (external drive).
9. Explain in detail how you access, setup, and implement system image backups (screen snips highly encouraged to prove you did the work):
10. Does this function exist on Windows 11? Is the process the same as on Windows 10 to create system images?
It still exists and is pretty much the same

Section 3: Setting Up System Restore Points

11. Goal: Configure and use System Restore to protect the operating system.
When there are issues after an update but data loss hasn't happened
12. Explain when to use System Restore and its limitations compared to system image backups.
13. *When there are issues after an update but data loss hasn't happened while system image backups are most for disaster situations.*
14. Manually create a system restore point.
15. Explain in detail how you access, setup, and implement restore points (screen snips highly encouraged to prove you did the work):
16. How do you test the restore point?
17. Does this function exist in Windows 11? Is the process the same as on Windows 10?
Yes it still exists in windows 11 and is mostly the same, however it adds a feature to delete points older than 60 days.

Section 4: Explain how you could automate backups in Windows 10 or Windows 11

Summary Questions:

1. What is the difference between File History, System Restore, and System Image backups?

File history backs up files to preserve older versions, system restore reverts the os and settings to older states, system image creates a complete image of the hard drive in case of a major failure.

2. When would you recommend creating a system image backup vs. a file backup?

System image backup would be better for when an entire system needs to be moved or restored while file backups are better for saving specific things.

3. What are the limitations of System Restore compared to other backup options?

It doesn't backup personal files and is less effective against some types of malware